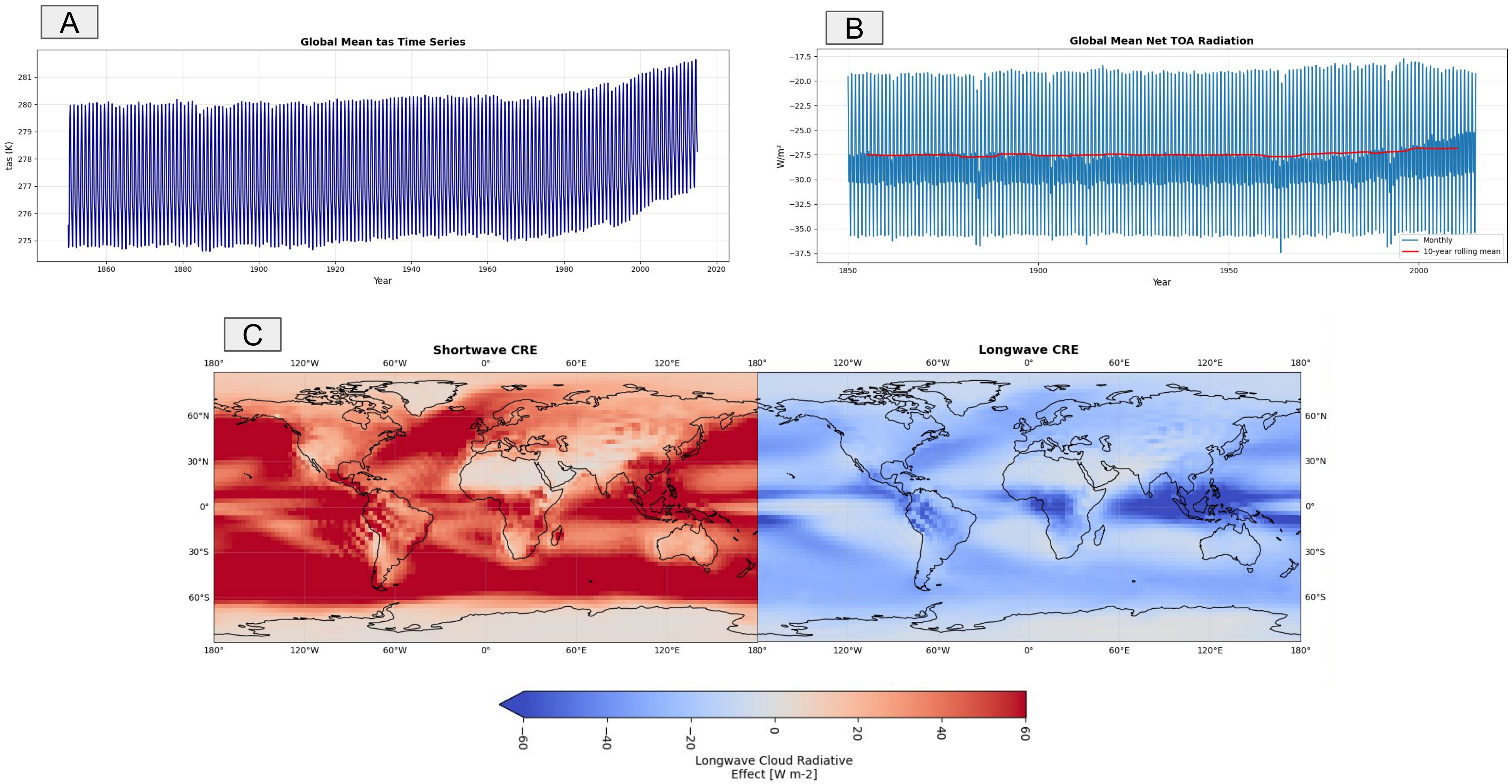

A
Global Mean tas Time Series
tas (K)
Year
B
Global Mean Net TOA Radiation
W/m²
Year
Monthly
10-year rolling mean
C
Shortwave CRE
Longwave CRE
Longwave Cloud Radiative Effect [W m-2]

Figure 1: Climate Trends and Cloud Radiative Forcing

(A) Global Mean Surface Air Temperature (tas) Time Series: Displays the evolution of global mean surface temperature dating back to 1850. This shows a clear positive trend especially post-1950. This demonstrates the primary input variable when it comes to understanding the sensitivity of the earth's climate. **(B) Global Mean Net Top-of-Atmosphere (TOA) Radiation:** Displays the monthly and 10-year rolling mean net radiation balance at the TOA. There is a upward trend in the recent years that are important for training models on the earth's energy balance. Patterns in both A and B suggest that a predictive model can be created. This is used to create the CRE datasets for the CNN model. **(C) Spatial Distribution of Cloud Radiative Effects (CRE):** Displays two global maps of shortwave (left) and longwave (right) CRE in W/m^2. The shortwave CRE map highlights the albedo effect being present in ocean regions. There are many clouds that reflect the shortwave radiation back into space which has a cooling effect on the earth's surface. The longwave CRE map shows the greenhouse effect of clouds across the tropic regions. The high altitude areas being near the equator pushes clouds up high and therefore colder. This allows them to "trap" heat more effectively, reducing the amount of longwave radiation reaching space. This demonstrates the output or results that the CNN model forms when given input temperatures.